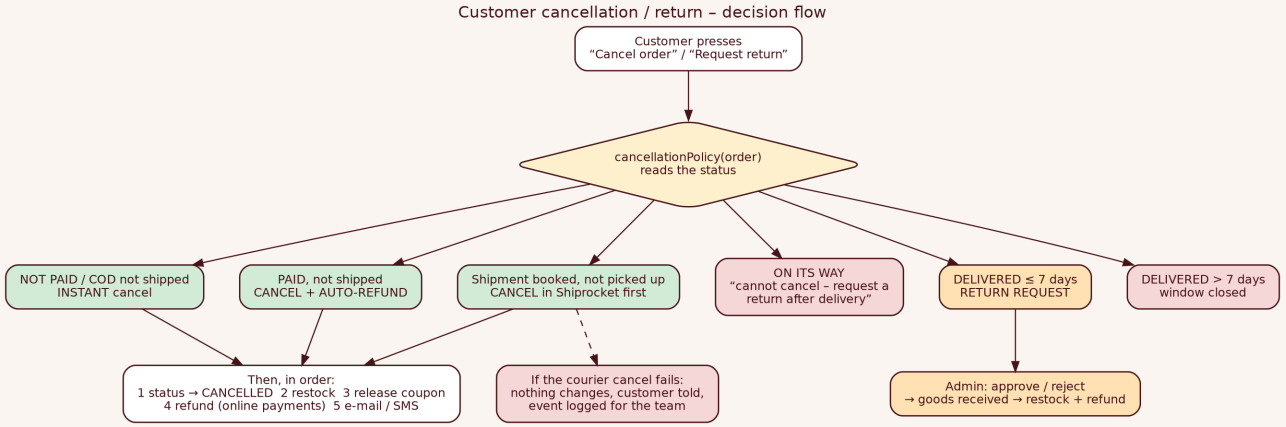
LUXE & THREAD · DOCUMENTATION

INTERNAL – for developers

Internal 13 · Order Cancellation & Returns – Design and Flow

Audience	INTERNAL – developers, support
Code	`lib/cancellation.ts` (rules) · `lib/orders/service.ts` (`cancelOrder`, `refundOrder`, `requestReturn`, `decideReturn`, `expireStaleOrders`) · `OrderActions.tsx` · `/api/orders/cancel` · admin order & returns pages



1 · Policy table (one source of truth: `cancellationPolicy`)

Order status	Mode	What the customer sees	What happens
Awaiting payment / payment failed	**instant**	"You can cancel this order now."	CANCELLED · restock · release coupon · no refund
COD confirmed (not shipped)	**instant**	same	same
Paid (not shipped)	**refund**	"…payment will be refunded to the original method"	CANCELLED · restock · coupon · **Razorpay refund (full)** · e-mail
Shipment created (not picked up)	**cancel-shipment**	"parcel not picked up – we can still cancel"	**Shiprocket cancel first** → then as above
On its way	none	"request a return after delivery"	–
Delivered ≤ 7 days	**return**	"request a return within 7 days"	RETURN_REQUESTED + admin workflow
Delivered > 7 days / other	none	window closed	–

Return window constant: `RETURN_WINDOW_DAYS = 7` (`cancellation.ts`).

2 · Order of operations (cancel)

1. Check policy (reject politely if not allowed).
2. If `cancel-shipment`: call Shiprocket cancel. **If it fails → stop**, keep the order unchanged, store `CANCEL_BLOCKED` event, tell the customer "our team will cancel it shortly" (no half-cancelled state).
3. `updateOrder` (row lock): status → CANCELLED via the state machine; store reason/time.
4. `restockOrder` – returns stock; **idempotent** (only once per order, via `stock_movements`).
5. Release coupon use.
6. If prepaid & captured → `refundOrder` (see §4). Refund failure is logged (`REFUND_FAILED`) and the order stays cancelled; owner retries.
7. Queue `order_cancelled` message; write `audit_log`.

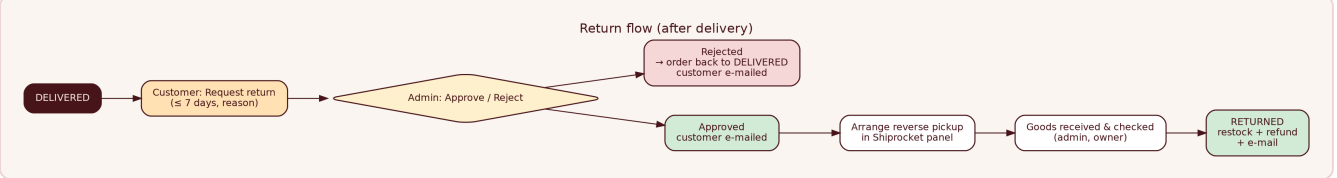
3 · Edge cases

Case	Handling
Customer abandons payment	cron expiry cancels after 30 min, restocks, releases coupon
Pays **after** expiry/cancel	`confirmPrepaidPayment` sees CANCELLED → records `LATE_PAYMENT` → auto-refund
Double click on Cancel	second call finds status CANCELLED → "can no longer be cancelled"
Cancel and webhook race	row lock + state machine → one wins; late "failed" cannot undo "paid"
Parcel picked up just now	Shiprocket rejects cancel → `CANCEL_BLOCKED`; owner handles as RTO/return
Partial refunds	owner can refund any amount ≤ remaining; status PARTIALLY_REFUNDED
COD order cancelled	no money; cancels shipment if booked
Refund to a closed account	Razorpay returns failure → manual follow-up

4 · Refund rules (`refundOrder`)

Only PREPAID with captured payment; amount default = remaining; must be `0 < amount ≤ total − refunded`; calls Razorpay refund API (`speed: normal`); records `refunds` row + order event + `refunded_amount`; payment status becomes REFUNDED or PARTIALLY_REFUNDED; sends `refund_processed`. **Owner role** required for manual refunds.

5 · Returns



1. Customer: order page → **Request a return** + reason → status `RETURN_REQUESTED`, row in `return_requests`.

2. Admin → Returns: **Approve** (e-mail) or **Reject** (order back to DELIVERED, e-mail with note).
3. After approval, create the **reverse pickup in Shiprocket** (manual) and wait for the parcel.
4. **Goods received → refund** (owner): status RETURNED · restock (`return_restock`) · refund (default full, or a partial amount if damaged) · return marked `refunded`. RTO (courier returns an undelivered parcel): status RTO from courier updates, stock restocked automatically; refund prepaid manually if agreed.

6 · Customer communication

Cancelled · refund processed · return approved/rejected – via the outbox (see Internal 12).

7 · Policy pages must match

`/policies/refund-and-cancellation-policy` and `shipping-policy` should state the same 7-day window and refund timelines. Change the constant and the text together.

8 · Tests

Unit: `cancellationPolicy` for every status, return window edges, state transitions. Provider checklist items 4, 8, 9, 14 in Internal 11.